

Document: Dewatering Method Statement
Reference: MS-1-P393D
Revision: 5



Project: 4B+G+10 Resi Bldg, P#JVC15AMRM004 (Stonehenge -2), JVC - Dubai

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Dewatering Method Statement

		 T.M.F. EURO FOUNDATIONS (LLC) تي.إم.إف. الأوروبية للأساسات (ش.ذ.م.م)	
Client	Consultant	Enabling Contractor	Main Contractor

5	13-10-23	Change in Deepwell Location	PM	AK
4	03-10-23	Change in Excavation Levels	PM	AK
3	13-10-23	Change in Excavation Levels	PM	AK
2	08-07-23	Discharge through Nakheel Line	PM	AK
1	19-06-23	Changes in Deepwell Locations as per the Consultant	PM	AK
0	09-06-23		PM	AK
Rev	Date	Description of Revision	Prepared By	Checked By

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Project: 4B+G+10 Resi Bldg, P#JVC15AMRM004 (Stonehenge -2), JVC - Dubai

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1.0 Purpose:

The purpose of this method statement is to describe the activities and procedures for the de-watering system installation & operation, to enable the construction works carried out in dry.

2.0 Scope:

The scope involves the de-watering of the proposed site to the required levels, to enable the construction works carried out in dry, based on the following details available.

Project	
Project	4B+G+10 Resi Bldg
Plot / Location	P#JVC15AMRM004(Stonehenge -2), JVC - Dubai
Dewatering Plot Dimensions	
Plot length	45 m
Plot width	39 m
Plot perimeter	168 m
Plot area	1,755 m ²
Levels	
Natural ground	+8.10 m DMD
Natural water	+6.25 m DMD
General excavation	- 6.10 m DMD
Maximum excavation	- 7.80 m DMD
Shoring Details	
Shoring type	Secant Pile On 4 Sides
Shoring toe	-9.30,10.00, &-11.65 m DMD
Discharge	
Discharge length	2400m from the site

3.0 Natural Ground Water:

Natural groundwater levels are considered as mentioned in section 2.0 based on the information provided at the time of dewatering system design. Kindly note that natural groundwater level might vary based on seasonal/tidal changes and/or dewatering activities in the vicinity of the proposed site.

4.0 References For Design:

Shoring/Excavation drawing	SH-00
Soil investigation report	SR/2209157
Natural groundwater level	Based on SIR

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5.0 Description Of Dewatering System:

We propose to install a Deepwell dewatering system to dewater the site as per the levels mentioned in section 2.0 above.

Deepwell dewatering system involves the installation of perforated UPVC casing & submersible pumps in the ground.

The proposed excavation shall be in two stages. The first stage of excavation shall involve the excavation to approximately 50 cm above the existing groundwater table.

Following the installation and operation of the dewatering system, the groundwater shall lower to approximately 0.5 m additional to the final excavation drawdown level to facilitate the second stage of excavation.

Note: General & additional drawdown may require French drains if ground conditions do not facilitate uniform drawdown or there are lateral seepages.

5.1 Control Of Seepages:

The presence of hard heterogeneous impermeable strata with varying horizontal & vertical permeability may not allow uniform drawdown curve formation (A theoretical drawdown curve can be presented; however, the strata may not necessarily allow the formation of such uniform drawdown profile).

In such ground conditions, "French Drain" is a suitable method to collect water from such strata & channel to the nearest deep wells. French Drains involves the installation of gravity drains (approx. fall of 1%, to ensure low water velocity & protection against erosion) which channel water into a sump/deep well.

Thus, if any seepages are encountered from the ground during excavation or at formation level, it is necessary for French drains (0.5m to 1m wide & deep) to collect this water into the nearest deep wells. The French drains shall be lowered in steps as the excavation progresses to reach the final level. Once the final excavation has been reached, a permanent French drain network (filled with suitable aggregate) to be installed based on site conditions.

Note: Plugging of leakages through shoring/anchors or ground cavities to be carried out by enabling/main contractor. Free flow of water through such sources into the site is not recommended as it may result in wash away of fines from outside the plot & may eventually result in settlements.

5.2 Dewatering System After Final Excavation:

After excavation to the required level is completed, (for deep wells which fall in the raft area), the discharge pipe & cable shall be countersunk 50 cm below the final excavation level (& covered with suitable aggregate) & be re-routed from the sides of Secant Pile On 4 Sides

5.3 Manhole Opening For Deepwells:

Based on the above, main contractor to provide an opening of 1.5m x 1.5m in the proposed raft at deep well location for pump repair & maintenance purposes. The raft openings design shall be as per the structural contractors design.

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6.0 Deepwell Installation:

Deep wells quantity

6 nos.

(Refer to drawing for details)

6.1 Deepwell Construction:

It is recommended that the construction of the temporary deep well system be carried out using a rotary drilling rig to install long temporary steel casings from the working platform level to prevent collapses during drilling. Use of bentonite or a similar method is not recommended as it may block the pores in the ground reducing well yield.

- 6.1.1 The main contractor shall provide a working platform not more than 0.5m above site water level.
- 6.1.2 Using an auger bucket the temporary casing can be cleaned of any drilling slurry or debris.
- 6.1.3 After cleaning out of the casing the slotted screen UPVC pipe shall be installed in the centre of the wellbore.
- 6.1.4 The annulus between the temporary casing and the UPVC deep well screen casing shall be filled with suitable aggregate to allow water drainage into the screen and prevent excessive fine material from passing through into the pumping area. A temporary cover shall be used to seal the top of the screen pipe to prevent accidental fall of debris.
- 6.1.5 On completion, the drilling rig shall be used to remove the temporary steel casing.

6.2 Deepwell Development:

Before to commissioning the pumping system, it is necessary to develop the deep well to ensure the safe and efficient performance of the well. Well development provides a means to improve the natural permeability of the ground around the well.

- 6.2.1 A rigid hydraulic pipe connected to a compressor is inserted into the well.
- 6.2.2 The well is filled artificially with water (if required) and compressed air is fed into the well using an air-lifting lance.
- 6.2.3 The compressed air will create a venturi effect which shall ensure the circulation of water through the filter material and into the well. This will ensure the removal of any dirt and fines in the vicinity of the well. This process shall be continued until the water emerging from the top of the well is free from fines.

6.3 Pump & Power Arrangement:

- 6.3.1 A centralized power source shall be provided to distribute power from Generators to various submersible pumps.
- 6.3.2 In the event of running Generator failure, the St-by Generator shall be started & put into service using change over switch.
- 6.3.3 The Control Panels shall send out an audio siren & visual alarm in case of any abnormal conditions such as failure of the submersible pump.

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6.3.4 The submersible pump shall receive power from the Control panels & shall be connected to a Distribution board, which shall receive power from the main supply generators.

6.4 Installation Of Submersible Pumps:

6.4.1 Suitable 3-phase Electrical submersible pumps shall be installed within each deep well.

6.4.2 The pumps shall be connected to discharge pipes, cables and ropes before to lowering.

6.4.3 A test shall be carried out on the pumps electrical operation before installation.

6.4.4 The units shall be lowered with the pipe & cable strapped to the lifting rope into the UPVC screen casing. The lifting rope shall be secured at the top of the well for removal of the pump for repair & maintenance.

6.5 Connection Of Pump To Discharge Line:

6.5.1 The final piping arrangement from the submersible pump shall be a rigid / semi-rigid pipe connected to the discharge header/discharge point, allowing the discharge to follow the profile of the excavation/shoring.

6.5.2 A non-return valve (N R V) shall be inserted at the connection point into the discharge where the pump is connected into a closed system. A check valve shall be connected downstream of the N R V to enable the operator to inspect the pump operation & control the flow of water.

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7.0 Discharge Control Plan:

- 7.1 The annulus between the deep well bore & UPVC deep well screen casing shall be filled with suitable aggregate to allow filtered drainage into the screen and prevent fine material from passing through into the pumping area.
- 7.2 Before to commissioning of the pumping system, it is necessary to develop the deep well, not only to ensure safe and efficient performance of the well, but also to wash out the dirt inside the well & within the aggregate filter pack.
- 7.3 The pumped water shall be feed into the discharge hose, which will be connected to the settlement tank. This tank shall have a baffle plate, which shall allow suspended solids, if any, to fall out of suspension and allow floating contaminants (diesel fuel etc.) to be isolated and removed using absorbent mats or skimming.
- 7.4 Discharge water shall be delivered only to the approved discharge location.
- 7.5 Allowance of 7 days for the water table to reduce after the dewatering system is started and before starting excavation.
- 7.6 Any voids/leakages in shoring to be immediately plugged upon discovery to prevent fines from being washed into the discharge system.
- 7.7 Deep Well UPVC casings not to be damaged during excavation.

Discharge requirements as per	Trakhees/Nakheel/RTA
Flow quantity at steady state	46.50 m ³ /hr (at the start can be twice)
Discharge location	Nakheel Line
Discharge water analysis	Weekly / Monthly as required
Water sample point	Sedimentation tank

8.0 Water Drawdown & Flow Monitoring:

Water Levels in deep wells shall be checked & recorded in the "Monitoring checklist" by the day shift and night shift watchman daily using a dip meter.

Water flow shall be measured using a standard V-notch tank, which shall also be recorded daily in the Monitoring checklist.

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9.0 Dewatering Shutdown & Decommissioning:

Once the requirement for the dewatering system ceases and based on the written instructions from the consultant/main contractor, the dewatering system can be switched off in the following sequence (based on the site conditions, sacrificial or non-sacrificial submersible pumps method shall be used) to facilitate the concreting & waterproofing of the 1.5m x 1.5m raft foundation openings (if provided).

The sequence of stopping the deep well shall be given before shutting down based on the final evaluation of the well recovery time. The sequence is determined by stopping selected well(s) in sequence to determine the time required for rise in water level.

If the water level is maintained below 2m of top of the well for 24 to 48 hours, the "Non-Sacrificial Well Method" shall be followed.

If the water in the well rises to within 2m from the top of the well, the pump needs to be kept operational & sacrificed using the "Sacrificial Well Method" to facilitate manhole water-proofing, steel, concreting & curing in dry.

9.1 Closing Steps for Sacrificial Well:

- 9.1.1 Put a cover plate on top of the deep well while keeping the pump running (by Dwex).
- 9.1.2 Do necessary waterproofing, blinding, raft works as per the main contractor's design in the manhole, keeping the submersible pump running (by Main Contractor).
- 9.1.3 Allow sufficient time for curing while keeping all pumps running (by Main Contractor).
- 9.1.4 After the curing time for the sacrificial deep wells has been completed, stop the non-sacrificial wells in sequence based on the written instructions from the consultant (while keeping the sacrificial pumps running to maintain a safe water level).

9.2 Closing Steps for Non-Sacrificial Wells:

- 9.2.1 Remove the Submersible Pump from the deep well (by Dwex).
- 9.2.2 Put a cover plate on top of the deep well (by Dwex).
- 9.2.3 Do necessary waterproofing, blinding, raft works as per the main contractor's design in the manhole (by Main Contractor).
- 9.2.4 Allow sufficient time for curing (by Main Contractor).
- 9.2.5 Complete dewatering system shut down based on written instructions from consultant/main contractor (By Dwex).

Note: Dwex is responsible for maintaining water level as per the scope for the duration of contractors dewatering requirements; however Dwex is not responsible for uplift calculations or for the dewatering stoppage recommendation. This must be proposed either by the Structural Designer/Design Consultant/Structural Contractor.

Shutting down of dewatering system may result in a rise in the water table to natural groundwater table & can be affected by the seasonal/tidal changes and/or dewatering activities in the vicinity of the site.

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10.0 Resources:

10.1 Responsibility Matrix:

Item	Dwex	Enabling / Main Contractor
Site Store & Office		✓
Discharge Permits & Fees		✓
RTA Fees & Deposits		✓
Craneage (mob/demob, on-site activities)		✓
4 Helpers To Assist Installation		✓
Deepwell Drilling		✓
Supply Of Aggregate For Deepwell/OW	✓	
Distribution Of Aggregate On-Site		✓
Supply Of Deepwell UPVC Screen	✓	
Protection For Pipes Crossing Road/Access		✓
Supply Of Fuel	✓	
Watchkeeping	✓	
Excavation Of Trenches For French Drains		✓
Supply Of Aggregate For French Drains		✓
Flow meter 6" as per Nakheel requirement		✓

10.2 Equipment:

Submersible pumps	Running pump(s) + 1 st-by
Genset	Running genset(s) + 1 st-by
Booster pump	As per site requirements
Compressor for deep well airlifting	As per site requirements

Refer to drawing for details

10.3 Material:

UPVC Screen pipes for deep wells	Refer to drawing for details
Permanent MS sleeve (in shoring)	
Pre-washed aggregate	
Layflat/PVC deep well discharge pipes	
GI/Layflat/PVC header system	

10.4 Manpower:

10.4.1 During installation:

Foreman	1 no.
Technician	1 no.
Electrician	1 no.
Skilled Labours	2 nos.

10.4.2 During operation & maintenance:

Day watchkeeping technician	1 no.
Night watchkeeping technician	1 no.
Back-up & emergency team	24 Hours on call

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Back-up & emergency number

+971 55 671 4100

11.0 Safety:

Work will be carried out in accordance with relevant Main Contractors safety documentation.

Special Safety Measures:

- All cables shall be well protected.
- Safe lifting procedures shall be followed.
- Safe hot-work procedures shall be followed.
- Personnel to wear safety shoes and protective helmets.
- Oil / Diesel leakage of pumps shall be rectified.
- Discharge water shall be delivered only to the approved discharge location.
- Housekeeping shall be maintained on-site in accordance with the Main Contractor's safety and hygiene standards.

12.0 Environmental Protection:

Environmental protection has been considered for the design of the proposed system.

12.1 Emission:

To limit emissions on the site, dewatering pump engines will be well maintained.

12.2 Oil & Diesel Spillage:

Suitable drip trays will be placed under all pumps and oil tanks to contain and leakages and spills. Bulk storage fuel tanks shall be suitably located. All tanks shall be bunded to contain and waste or spillage.

12.3 Discharge Water:

The pumped water shall be feed into the discharge hose, which will be connected to the sedimentation tank. This tank shall have a baffle plate, which shall allow suspended solids to fall out of suspension and allow floating contaminants (diesel fuel etc.) to be isolated and removed using absorbent mats or skimming.

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13.0 Hazard & Risk Assessment:

To evaluate risk:

Likelihood (L) X Severity (S) = Risk (R)

Where Risk is termed:

Low	1 to 5
Medium	6 to 10
High	11 to 25

Risk Rating

L = Likelihood	S = Severity
1 Very Unlikely	1 Minor Injury
2 Unlikely	2 Lost Time Injury
3 Likely	3 Major Injury
4 Very Likely	4 Single Fatality
5 Certain	5 Multiple Fatalities

Activity / Element	Potential Hazards	Population at Risk	Risk Rating			Control Measures Specification	Residual Risk		
			L	S	R		L	S	R
General	Other contractors	Operatives	2	1	2	Attend site safety induction Comply with site rules. Co-ordination and sequencing of work to prevent a clash of activities Wear appropriate PPE.	1	1	1
General	Plant and equipment	Operatives	2	1	2	All plant and equipment to be suitably serviced and maintained by competent fitters.	1	1	1
General	Lifting gear & loads	Operatives	3	3	9	Check sling, ropes, and shackles for damage and capacity (certificates should be available for scrutiny). Only competent plant operators to use lifting devices. Slung loads to be secure and balanced, use guide ropes use hard standing for reception. Keep clear of slug loads; keep hands away from loads especially at the point of landing. Use qualified banks man. Maintain communication between banks man and drivers.	1	3	3
Completed bores	Trip hazards	Operatives Visitors	4	2	8	Completed bores to be backfilled as soon as possible. Open bores to have barriers or bunting around to make them more visible.	1	2	2
Layout equipment	Muscular-skeletal injury	Operatives	3	3	9	Wear suitable PPE. Tie loose clothing. Employ correct lifting techniques. Obtain assistance with loads if required. Use mechanical lifting gear if possible.	1	3	3
Electrical installation	Electric shock	Operatives	4	4	16	To be carried out by a qualified electrician only.	1	4	4
Maintenance	Trip Hazards	Operatives Visitors	2	2	4	Keep the site in a clean and tidy state.	1	2	2

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	Falling objects					Remove unwanted or unused items as soon as possible. Lay down hoisted equipment. Remove items from around the side of excavations.			
Maintenance	Falls into Excavations	Operatives Visitors	2	2	4	Keep away from excavation sides. Do not go on-site during the hours of darkness unless adequate lighting is available and only if accompanied by another person.	1	2	2
Groundwater	Wells disease chemical contamination	Operatives	3	3	9	Do not wash with groundwater. Wear appropriate PPE. Wash in clean water before eating, drinking, or smoking. Cover cuts and abrasions with waterproof plasters/dressings. Notify GP if flu-like symptoms occur.	1	3	3
Outdoor working	Sunburn	Operatives	5	2	10	Wear appropriate clothing. Apply sunblock to exposed skin.	1	2	2

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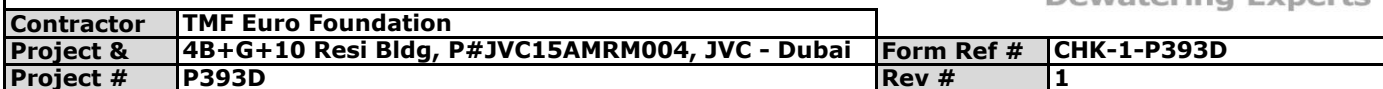


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APPENDIX A: MONITORING CHECKLIST

TO BE FILLED BY WATCHMAN



Genset

Booster Pump

Deepwell (*Datum as per site ref level)

Discharge System

Total Flow

WATCHMAN'S SIGN	FOREMAN / SITE ENG.'S SIGN
-----------------	----------------------------

Remarks :

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APPENDIX B: DEWATERING & DISCHARGE DRAWING

KEY PLAN :

NOTES & LEGEND :

	8" UPVC DEEPWELL
	2" Ø DISCHARGE PIPE
	6" Ø HEADER PIPE
	6" Ø DISCHARGE / LAYFLAT HOSE
	PROP. FRENCH DRAIN (0.50X0.50m)
	SEDIMENTATION TANK
	GENERATOR
	CONTROL PANEL
	SECANT PILE TOE: -9.30,10.00,-11.65m DMD

EXCAVATION LEVELS:

Level 01	-3.70m DMD
Level 02	-3.90m DMD
Level 03	-4.40m DMD
Level 04	-4.60m DMD
Level 05	-6.10m DMD
Level 06	-6.40m DMD
Level 07	-7.80m DMD
Level 08	-8.00m DMD

07	13/11/23	Change in Deepwell Location	PM	AK	MD
06	10/11/23	Change in Deepwell Location	PM	AK	MD
05	10/11/23	Change in Excavation level	PM	AK	MD
04	03/11/23	Change in Excavation level	PM	AK	MD
03	13/10/23	Change in Excavation levels	PM	AK	MD
02	08/07/23	Discharge through Nakhheel Line	PM	AK	MD
01	19/06/23	Change in Deepwell location as per Consultant	PM	AK	MD
00	09/06/23		PM	AK	MD
Rev.	Date	Description	Drawn by	Checked	Approved

CLIENT :



PROJECT TITLE :

4B+G+10 Resi Bldg, P#JVC15AMRM004, JVC, Dubai.

CONSULTANT :



SHORING CONTRACTOR :



SUB CONTRACTOR :



Drawing Title :

DEWATERING LAYOUT DRAWING

PROJECT NO: P393D	SCALE: N.T.S.	DATE: 13/11/23
DRAWN: PM	CHECKED: AK	APPROVED: MD
Dwg. No. DEW-1-P393D	Rev. 07	

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E 486976.064
N 2772616.704

E 486975.639
N 2772557.378

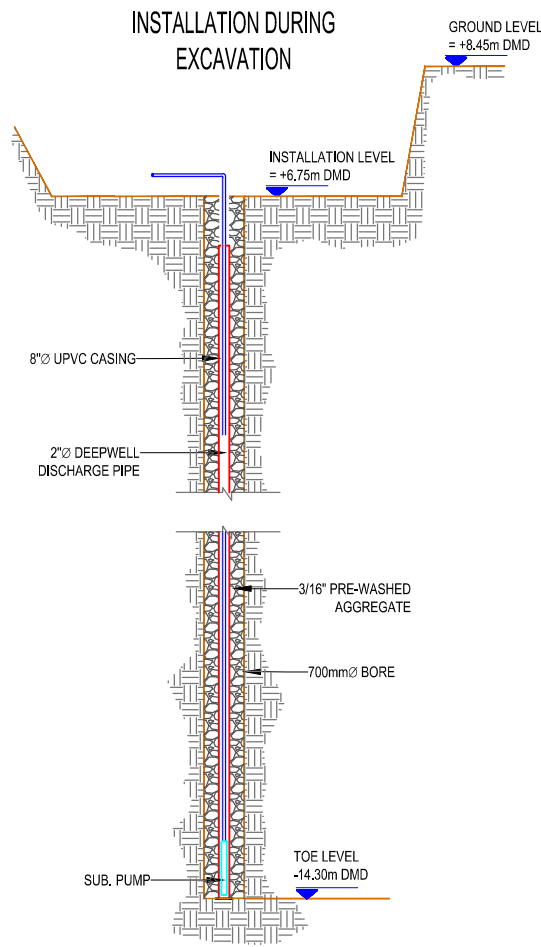
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DEEP WELL CO-ORDINATES

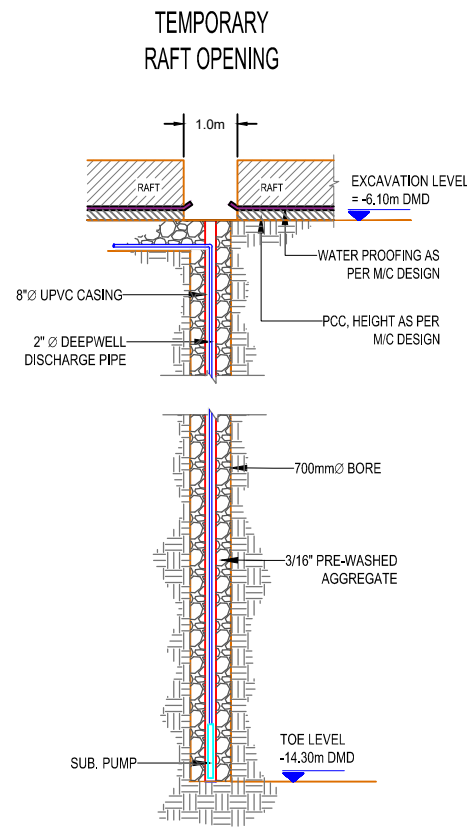
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3	2772580.00	486958.22
4	2772596.7	486976.34
5	2772608.33	486977.20
6	2772591.11	486994.8

EMPTY NEIGHBOR PLOT

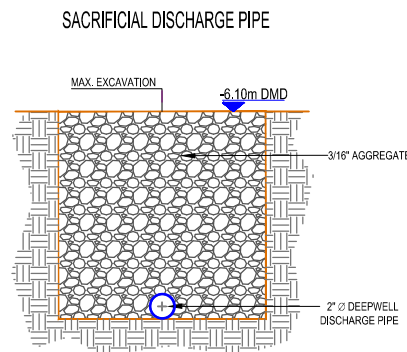
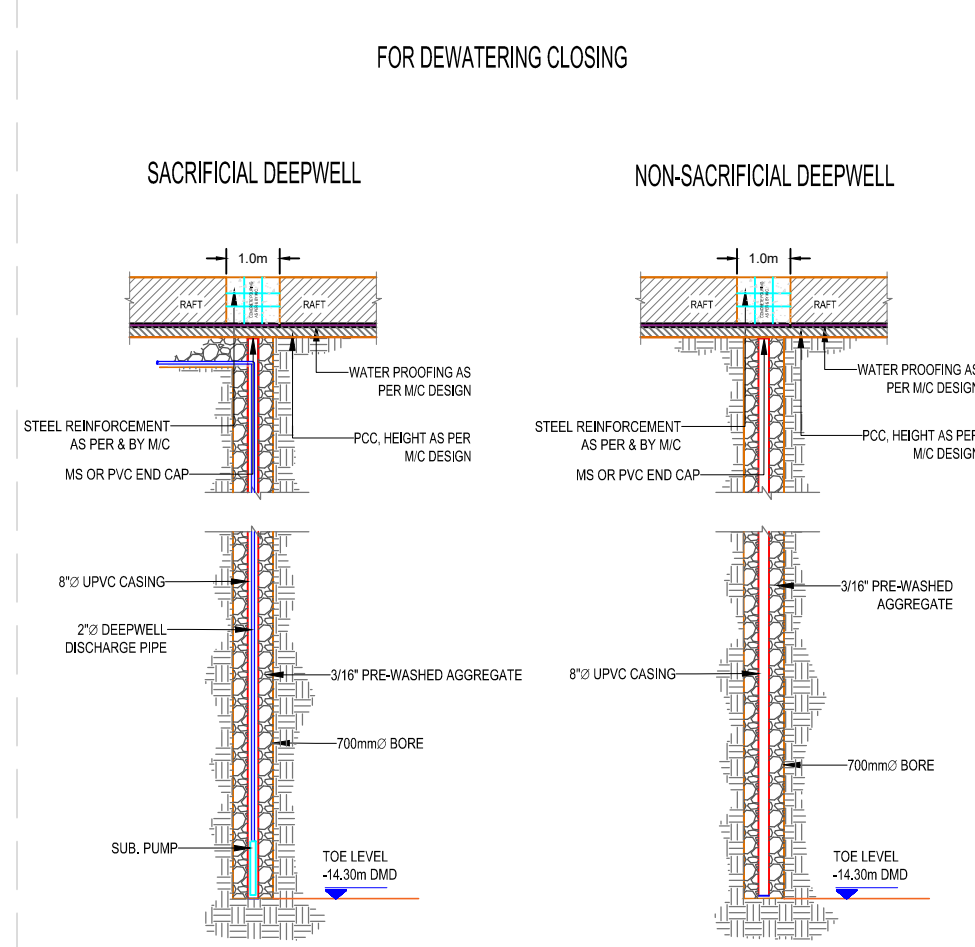
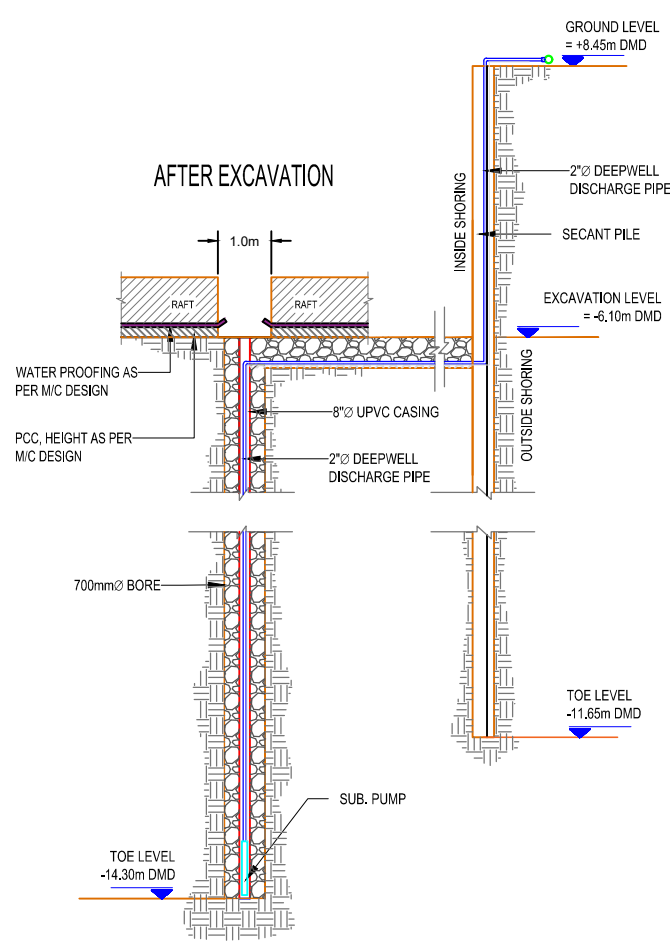
DISCHARGE TO THE
NAKHEEL LINE



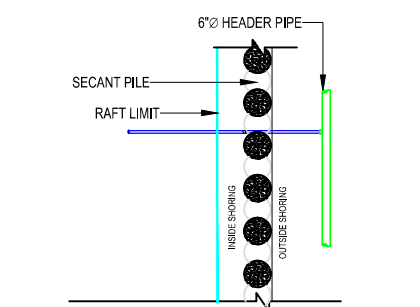
SECTION 1-1
DURING EXCAVATION



SECTION 1-1
AFTER EXCAVATION



SECTION 3-3



TOP VIEW BETWEEN SECANT PILE SHORING &
6"Ø HEADER PIPE

KEY PLAN :

NOTES & LEGEND :

05	10/11/23	Change in Excavation level	PM	AK	MD
04	03/11/23	Change in Excavation level	PM	AK	MD
03	13/10/23	Change in Excavation levels	PM	AK	MD
02	08/07/23	Discharge through header line	PM	AK	MD
01	19/06/23	Change in Deepwell location as per Consultant	PM	AK	MD
00	09/06/23		PM	AK	MD
Rev.	Date	Description	Drawn by	Checked	Approved

CLIENT :



PROJECT TITLE :

4B+G+10 Resi Bldg, P#JVC15AMRM004,
JVC, Dubai.

CONSULTANT :



SHORING CONTRACTOR :



SUB CONTRACTOR :



Drawing Title :

SECTION LAYOUT DRAWING

PROJECT NO: P393D	SCALE: N.T.S.	DATE: 10/11/23
DRAWN: PM	CHECKED: AK	APPROVED: MD
Dwg. No. SEC-1-P393D		Rev. 5

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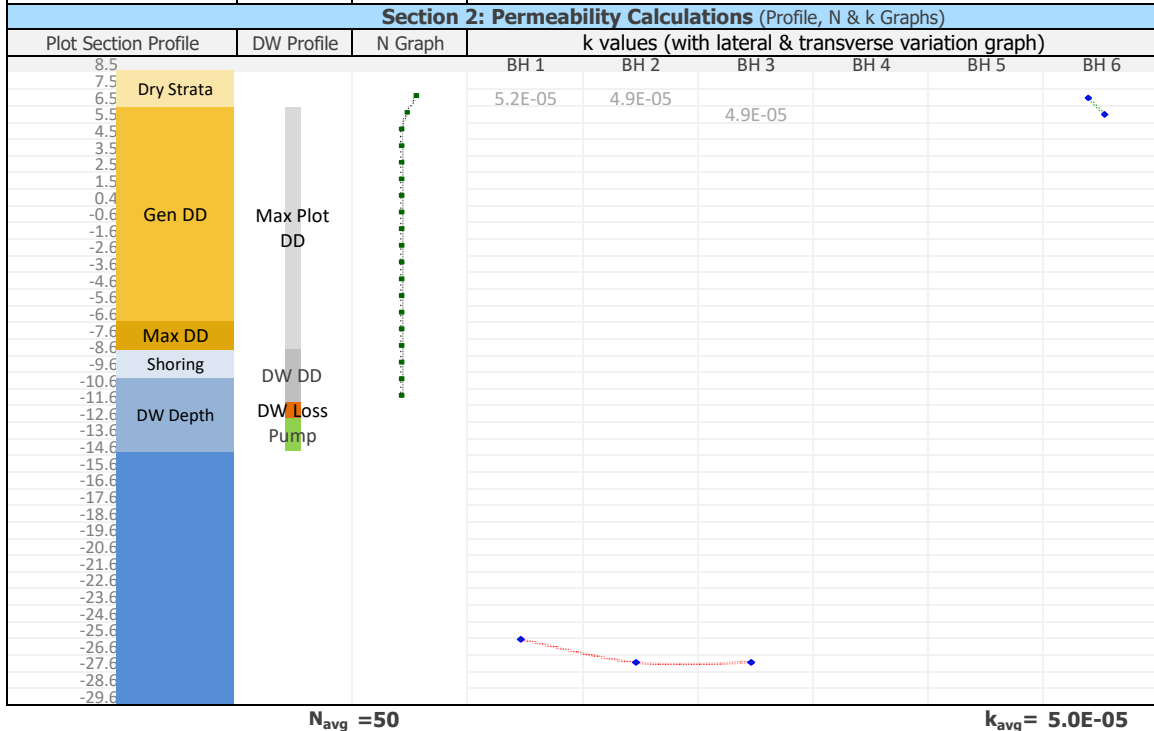
APPENDIX C: DEWATERING DESIGN CALCULATIONS

M/S TMF Euro Foundation
 Project 4B+G+10 Resi Bldg (Stonehenge -2)
 Plot / Loc: JVC15AMRM004, JVC - Dubai
 Subject Dewatering Design Calculations

Our Ref CAL-1-P393D
 Revision 3
 Date 03-Nov-23
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Section 1: Dewatering & Discharge Scope			
Dewatering Requirement		Discharge System	
Duration	150 Days	Discharge Location	Nakheel Line
Plot Length	45.00 m	Disch. Lm From Site	2,300.0 Lm
Plot Width	39.00 m	Elevation of Disch Point	2.0 m
Levels		Outfall Quantity	-
Natural Ground	8.45 m DMD	Water Analysis Every	3.0 months
Installation	6.75 m DMD	Disch. to Nakheel Line	☒
Natural Water	6.25 m DMD	Injection Pump	☒
Gen Excavation	-6.10 m DMD	References	
Max Excavation	-7.80 m DMD	Drawing Ref #	ST-100
Max Excavation Area %	15.00 %	SIR Ref #	SR/2209157
Drawdown			
Factor of Safety	0.50 m		
Gen Drawdown + FoS	12.85 m		
Max Drawdown + FoS	14.55 m		
Deepwell			
DW Bore Ø	700.0 mm		
DW Depth	21.2 m		
DW Depth Below M Exc	6.7 m		
DW Toe Level	-14.5 m DMD		
Shoring Type (by Main Cont.)			
Secant Pile On 4 Sides			
Shoring Toe	-10.00 m DMD		
MS Sleeve in Shoring ☐	N/A		
MS Sleeve Ø			

Remarks: -9.30, -10.00 & 11.65DMD (SH-01)



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M/S TMF Euro Foundation
Project 4B+G+10 Resi Bldg (Stonehenge -2)
Plot / Loc JVC15AMRM004, JVC - Dubai
Subject Dewatering Design Calculations

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Section 3: Dewatering Calculations: Equivalent Radius Method (CIRIA Report)		
a	Length of excavation	45.0 m
b	Width of excavation	39.0 m
a.b	Area	1,755.0 m ²
2.(a+b)	Perimeter	168.0 Lm
r _e	Equivalent well radius	23.6 m
W _L	Natural ground water level	6.25 m DMD
G _E	General excavation level	-6.10 m DMD
M _E	Maximum excavation level	-7.80 m DMD
H	Height of water from lower impermeable layer to natural water level	14.55 m
h _n	Average Drawdown (Gen & Max) required in equivalent well (0.5m FoS)	13.11 m
h _w	Height of water from lower impermeable layer to avg. drawdown level	1.45 m
k	Acuifer permeability	4.99E-05 m/s
R _o	Radius of influence	$C(H-h_w)\sqrt{k}$ m
Q	Steady state seepage/flow	$kn(H^2-h_w^2)/\ln(R_o/r_e)$ m ³ /s

Step 1: Radius of Influence

$$R_o = \frac{C(H-h_w)\sqrt{k}}{277.9} \text{ m (Calculated)}$$

$$R_c = R_o + r_e = 301.5 \text{ m}$$

Step 2: Steady State Seepage/Flow

$$Q = \frac{kn(H^2-h_w^2)}{\ln(R_c/r_e)} = 12.9 \text{ l/s} = 46.5 \text{ m}^3/\text{hr}$$

During initial drawdown, flow can be twice that of steady state.

Step 3: Deepwell Yield

q	Individual deepwell yield	nDL_wki m ³ /s
L _w	Wetted length of deepwell	4.68 m
D	Deepwell bore ø	700 mm
k	Acuifer permeability	5.0E-05 m/s
i	DW entry hydraulic gradient = 6 if $k < 2.3E-4$ OR $= 1/(15\sqrt{k})$ if $2.3E-4 < k < 2.8E-3$	6.00 units

$$q = nDL_wki = 3.1 \text{ l/s} = 11.1 \text{ m}^3/\text{hr}$$

Step 4: Number Of Deepwells

$$Q = 46.5 \text{ m}^3/\text{hr} \text{ (Total seepage in equivalent well)}$$

$$q = 11.1 \text{ m}^3/\text{hr} \text{ (Yield per deepwell)}$$

$$\text{Total DWs} = \frac{Q}{q}(1.2) = 6.0 \text{ Deepwells (With 20\% FoS)}$$